

opens the panel door.

Antenna for GSM modem. As the whole system is installed inside a metal box, there is a strong need to boost the GSM signals with adequately matching antenna. Many GSM modems available in the market provide antenna by just winding a coil of 14/36 SWG wire without appropriate connectors. Due to this, 80 per cent of the transmitted and received signal power is wasted or leaked. To provide strong signals, irrespective of low GSM signals in the rural area, a 50-ohm matching antenna and circuitry is incorporated in the antenna stage.

Layout and operation

Nano Ganesh-GSM is a wireless controller that receives the controlling demands from the farmer's mobile phones.

Ostwal says, "The Nano Ganesh-GSM unit comprises state-of-the-art life-tested technology to provide a feedback of power or load on/off status with the help of simple audio tones, hence making the system linguistically free."

The GSM unit easily connects to the starter panel with one of its wires connected to the starter input and the other used to sense the status of the load. It also has a neutral or an earth wire that must be connected to the GSM unit in order to generate single-phase, 230V power supply.

Before starting the pump, a farmer needs to check the availability of the power at the pump by m-connection. Thus, with the help of his mobile phone (the same mobile phone that he uses in his daily life), he connects with the GSM unit and, if there is adequate power, he hears a simple audio tone of 400Hz on his phone.

In case adequate power is not there, no tone is heard. Once status of the power supply is confirmed, he presses the 'start code' digits on his mobile phone, which sends DTMF tones to the GSM unit. GSM unit decodes the DTMF tones and switches on the electromechanical relay inside the unit. A motor-on relay triggers the auto switch to start the water pump.

Auto switch ensures proper start of the water pump by taking care of the phase sequence and single phasing.

As soon as water pump starts, a feedback is sensed at motor-on terminal of the GSM unit and the farmer hears a vibrating tone of the running motor and switches his phone off. The task is completed within a few seconds.

In case of any sudden failure of power while the pump is running, auto switch immediately stops the water pump. As soon as adequate power is resumed at the pump end, an alert call or SMS is again sent to the farmer's mobile phone.

Important point to note is that, if anyone opens the door of the panel, an alert or SMS is sent to prevent any theft or damage to the system. Hence, due to multiple functions of m-connectivity with the water pump, a farmer can monitor pump operations 24x7 from any distance without physically going to the pump area. ●

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